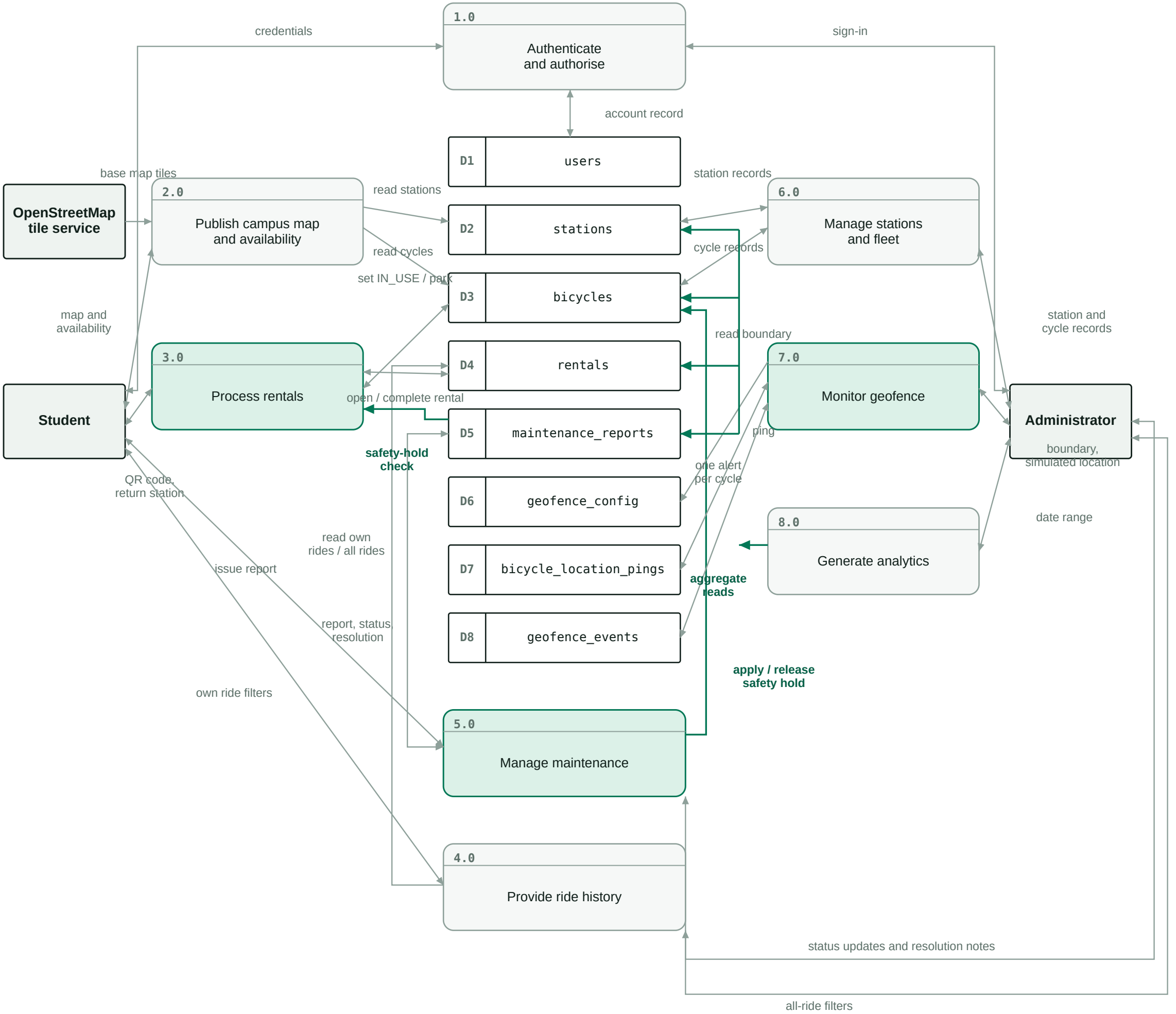
1.0
Authenticate and authorise
credentials
sign-in
account record
D1
users
base map tiles
OpenStreetMap tile service
2.0
Publish campus map and availability
read stations
read cycles
set IN_USE / park
D2
stations
D3
bicycles
D4
rentals
D5
maintenance_reports
D6
geofence_config
D7
bicycle_location_pings
D8
geofence_events
station records
6.0
Manage stations and fleet
cycle records
map and availability
station and cycle records
read boundary
3.0
Process rentals
open / complete rental
7.0
Monitor geofence
ping
Student
Administrator
boundary, simulated location
safety-hold check
one alert per cycle
QR code, return station
read own rides / all rides
issue report
report, status, resolution
own ride filters
8.0
Generate analytics
date range
aggregate reads
apply / release safety hold
5.0
Manage maintenance
4.0
Provide ride history
status updates and resolution notes
all-ride filters